

开源软件仓库与开源软件生态

概念与历史

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Outline

① 开源软件仓库

② 开源生态系统

软件仓库

定义

A software repository is a storage location from which software packages may be retrieved and installed on a computer.

Many software publishers and other organizations maintain servers on the Internet for this purpose, either free of charge or for a subscription fee. Repositories may be solely for particular programs, such as CPAN for the Perl programming language, or for an entire operating system^a.

^ahttps://en.wikipedia.org/wiki/Software_repository

典型软件仓库

软件仓库分类

- 操作系统发行版：基于某种内核的操作系统及外围软件构成的完整系统，如 Debian、Fedora 等 GNU/Linux 发行版，或 FreeBSD、OpenBSD 等 BSD 发行版
- 特定语言仓库：提供针对某个语言的软件包/类库等，如 Perl 的 CPAN、R 的 CRAN、 T_E X 的 CTAN、C++ 的 boost 等

软件仓库

软件仓库的操作

Operators of such repositories typically provide a package management system, tools intended to search for, install and otherwise manipulate software packages from the repositories. For example, many Linux distributions use Advanced Packaging Tool (APT), commonly found in Debian based distributions, or yum found in Red Hat based distributions. There are also multiple independent package management systems, such as pacman, used in Arch Linux and equo, found in Sabayon Linux.

APT 梗

今天就能开始用的小知识

```
$apt moo
```

```

          (__)
          (oo)
        /-----\
       / |         ||
      *  /\----/\
         ~~      ~~

```

```
... "Have you mooed today?" ...
```

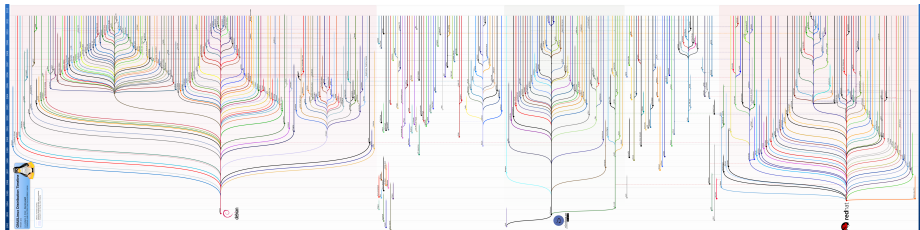
```
$ LC_ALL=C apt |tail -n 1
```

```
$ apt |tail -n 1
```

This APT has Super Cow Powers.

本 APT 具有超级牛力。

Linux 发行版



典型 Linux 发行版

Debian

Debian 是完全由自由软件组成的类 UNIX 操作系统，其包含的多数软件使用 GNU 通用公共许可协议授权，并由 Debian 计划的参与者组成团队对其进行打包、开发与维护。

Debian 项目最初由伊恩·默多克于 1993 年发起，Debian 0.01 版在 1993 年 9 月 15 日发布，而其第一个稳定版本则在 1996 年发布。^a

^a<https://zh.wikipedia.org/zh-cn/Debian>



典型 Linux 发行版

Debian 的特色

Debian 以其坚守 Unix 和自由软件的精神，以及其给予用户的众多选择而闻名。现时 Debian 提供了超过 25,000 个软件，超过 50,000 个软件包，并正式支持 10 个计算机系统结构。

作为一个大的系统组织框架，Debian 旗下有多种不同操作系统核心的分支计划，主要为采用 Linux 核心的 Debian GNU/Linux 系统，其他还有采用 GNU Hurd 核心的 Debian GNU/Hurd 系统、采用 FreeBSD 核心的 Debian GNU/kFreeBSD 系统等。众多知名的 Linux 发行版，例如 Ubuntu、Knoppix 和 Deepin，也都建基于 Debian GNU/Linux。

典型 Linux 发行版

今天就能开始用的小知识

- Debian 名字来源于发起人 Ian Murdock 和当时的女友 Debra Lynn
- Debian 的版本号昵称来自玩具总动员
- Debian 的 logo 来自玩具总动员角色巴斯光年的胡子 (存疑)^a

^aMartin F Krafft. *The Debian system: concepts and techniques*. No Starch Press, 2005.



典型 Linux 发行版

Ubuntu



典型 Linux 发行版

ArchLinux



典型 Linux 发行版

Red Hat



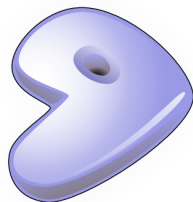
典型 Linux 发行版

Fedora



典型 Linux 发行版

Gentoo



gentoo linux

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所谓全栈

Full Stack

To me, a Full Stack Developer is someone with familiarity in each layer, if not mastery in many and a genuine interest in all software technology.

Good developers who are familiar with the entire stack know how to make life easier for those around them. This is why I'm so against silos in the work place. Sure, politics and communication challenges get in the way in large organizations. I think the point Facebook is going for with their hiring policy is, if smart people use their heads and their hearts, a better product gets built in less time^a.

^a<https://www.laurencegellert.com/2012/08/what-is-a-full-stack-developer/>

所谓全栈

Layers of the full stack

- Server, Network, and Hosting Environment.
- Data Modeling
- Business Logic
- API layer/Action layer/MVC
- User Interface
- User Experience
- Understanding what the customer and the business need

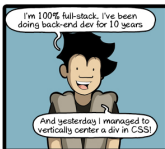
所谓全栈

Which full-stack developer are you?

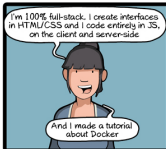
The Full-Stack Devops Engineer



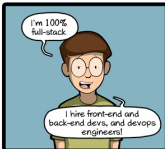
The Full-Stack Back-End developer



The Front-End Full-Stack Developer



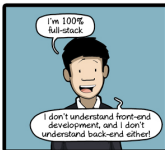
The Full-Stack CTO



The Full-Stack Intern



The Full-Stack Project Manager



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